

The Complete Two-Weight $\mathbb{Z}/5\mathbb{Z}$ Goemans–Linial Cone

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Abstract

For the weighted Cayley graph on $\mathbb{Z}/5\mathbb{Z}$ whose difference- ± 1 edges have weight a and difference- ± 2 edges have weight b , we determine the uniform-demand sparsest cut and the Goemans–Linial relaxation for every $a, b \geq 0$ with $a + b > 0$. Both values are

$$\frac{5 \min\{a + 2b, 2a + b\}}{6(a + b)}.$$

We classify every optimal cut up to dihedral symmetry and every optimal translation-invariant feasible semimetric, including both one-weight faces and the tie $a = b$. The key reduction is an objective-preserving translation average realized in an orthogonal direct sum. Fourier diagonalization then shows that the full invariant Goemans–Linial cone is the elementary two-distance triangle cone.

1 Setup and result

Let $\Gamma = \mathbb{Z}/5\mathbb{Z}$. The graph $G_{a,b}$ has the five unordered edges $\{x, x + 1\}$ of weight a and the five unordered edges $\{x, x + 2\}$ of weight b . Put $S = a + b > 0$. Equivalently, its normalized random walk uses the step distribution

$$p(1) = p(-1) = \frac{a}{2S}, \quad p(2) = p(-2) = \frac{b}{2S}.$$

For a nonempty proper subset $A \subset \Gamma$, with x, y independent and uniform in Γ and s sampled from this distribution, set

$$N_{a,b}(A) = \Pr[\mathbf{1}_A(x) \neq \mathbf{1}_A(x + s)], \quad D(A) = \Pr[\mathbf{1}_A(x) \neq \mathbf{1}_A(y)]$$

and $\psi_{G_{a,b}}(A) = N_{a,b}(A)/D(A)$. We write $\psi(G_{a,b})$ for the minimum over such A .

A Goemans–Linial feasible semimetric is a function $d(x, y) = \|p_x - p_y\|_2^2$ that also obeys every triangle inequality. It is permitted to vanish on distinct points. Define

$$N_{a,b}(d) = \mathbb{E}_{x,s} d(x, x + s), \quad D(d) = \mathbb{E}_{x,y} d(x, y),$$

and, in the random-walk/uniform-demand normalization of Stamoulis [1],

$$\text{SDP}_{\text{GL}}(G_{a,b}) = \inf_{\substack{d \text{ GL-feasible} \\ D(d) > 0}} \frac{N_{a,b}(d)}{D(d)}.$$

We regard cuts as unordered bipartitions, so A and A^c are identified. The dihedral group D_5 acts by $x \mapsto \pm x + c$.

Theorem 1. For all $a, b \geq 0$ with $a + b > 0$,

$$\text{SDP}_{\text{GL}}(G_{a,b}) = \psi(G_{a,b}) = \frac{5 \min\{a + 2b, 2a + b\}}{6(a + b)}. \quad (1)$$

The optimal cuts, up to D_5 , are exactly as follows:

- if $a > b$ (including $b = 0$), the adjacent-pair orbit represented by $\{0, 1\}$;
- if $b > a$ (including $a = 0$), the difference-two-pair orbit represented by $\{0, 2\}$;
- if $a = b > 0$, every nontrivial cut.

For a translation-invariant semimetric put

$$u = d(0, 1) = d(0, 4), \quad v = d(0, 2) = d(0, 3).$$

Modulo positive scaling, the nonzero optimal invariant feasible semimetrics are exactly the ray $v = 2u$ when $a > b$, the ray $u = 2v$ when $b > a$, and every ray with $u, v > 0$, $u \leq 2v$, and $v \leq 2u$ when $a = b > 0$. Moreover, translation averaging preserves feasibility and the objective, so no non-translation-invariant feasible semimetric gives a smaller value.

The nearest result in the cited source is the exactness statement for ordinary cycles, which covers the two coordinate faces after relabeling [1, Proposition 6.1]. Its character-based exactness theorem assumes that a minimizing character has image size at most four, while every nontrivial character of $\mathbb{Z}/5\mathbb{Z}$ has image size five; hence that hypothesis does not itself cover the interior two-weight cone [1, Theorems 1.1 and 4.6]. The comparison here is limited to that cited version and is not a claim of priority.

2 Cuts and translation symmetrization

For $i = 1, 2$, let $c_i(A)$ be the number of crossing unordered edges in the difference- i family. If $k = |A|$, counting the two orientations of every crossing edge gives

$$N_{a,b}(A) = \frac{ac_1(A) + bc_2(A)}{5S}, \quad D(A) = \frac{2k(5 - k)}{25},$$

and consequently

$$\psi_{G_{a,b}}(A) = \frac{5(ac_1(A) + bc_2(A))}{2Sk(5 - k)}. \quad (2)$$

Complementation reduces the enumeration to $k \leq 2$. There is one orbit of singletons and two orbits of pairs, represented by $\{0\}$, $\{0, 1\}$, and $\{0, 2\}$. Direct counting yields

A	(c_1, c_2)	$\psi_{G_{a,b}}(A)$
$\{0\}$	$(2, 2)$	$5/4$
$\{0, 1\}$	$(2, 4)$	$5(a + 2b)/(6S)$
$\{0, 2\}$	$(4, 2)$	$5(2a + b)/(6S)$

(3)

The second entry is strictly least exactly when $a > b$, the third is strictly least exactly when $b > a$, and all three agree exactly when $a = b > 0$. This proves the cut value and its complete equality classification.

We next justify the symmetry reduction for the relaxation.

Lemma 2 (Lossless translation averaging). *For every GL-feasible d on Γ , the function*

$$\bar{d}(x, y) = \frac{1}{5} \sum_{c \in \Gamma} d(x + c, y + c) \quad (4)$$

is translation-invariant and GL-feasible, and satisfies $N_{a,b}(\bar{d}) = N_{a,b}(d)$ and $D(\bar{d}) = D(d)$.

Proof. Write $d(x, y) = \|p_x - p_y\|^2$. In the orthogonal direct sum of five copies of the ambient Euclidean space, define

$$q_x = 5^{-1/2} (p_{x+c})_{c \in \Gamma}.$$

Then $\|q_x - q_y\|^2 = \bar{d}(x, y)$, proving squared-Euclidean realizability. Averaging the five translated instances of $d(x, z) \leq d(x, y) + d(y, z)$ proves the corresponding triangle inequality for $\bar{d}$. Translation invariance follows by reindexing c . Finally, translation invariance of the uniform finite sums gives

$$\mathbb{E}_{x,s,c} d(x + c, x + s + c) = \mathbb{E}_{x,s} d(x, x + s), \quad \mathbb{E}_{x,y,c} d(x + c, y + c) = \mathbb{E}_{x,y} d(x, y).$$

Thus both objective terms are unchanged. In particular a positive denominator remains positive. $\square$

Invariant metrics are a subclass of all feasible metrics, while Lemma 2 maps every member of the larger class into that subclass without changing its ratio. The two infima therefore coincide.

3 The invariant cone

An invariant symmetric hollow distance matrix has first row

$$(0, u, v, v, u). \quad (5)$$

Recall that a finite symmetric hollow matrix M is a squared-Euclidean distance matrix if and only if it is conditionally negative semidefinite:

$$z^\top M z \leq 0 \quad \text{whenever} \quad \sum_x z_x = 0. \quad (6)$$

For completeness, if $M_{xy} = \|p_x - p_y\|^2$, then the left side of (6) is $-2\|\sum_x z_x p_x\|^2$. Conversely, (6) makes $-\frac{1}{2}JMJ$ positive semidefinite for $J = I - \frac{1}{5}\mathbf{1}\mathbf{1}^\top$; a Gram factorization of this matrix realizes the distances M_{xy} .

Let $\omega = e^{2\pi i/5}$. The nonconstant Fourier vectors diagonalize (5), with eigenvalues

$$\lambda_k = 2u \cos(2\pi k/5) + 2v \cos(4\pi k/5), \quad k = 1, 2, 3, 4.$$

Writing $\varphi = (1 + \sqrt{5})/2$ and using $2 \cos(2\pi/5) = 1/\varphi$ and $2 \cos(4\pi/5) = -\varphi$, these are

$$\lambda_1 = \lambda_4 = u/\varphi - \varphi v, \quad \lambda_2 = \lambda_3 = -\varphi u + v/\varphi.$$

The four nonconstant Fourier vectors form a basis of the zero-sum subspace. Thus the exact squared-Euclidean conditions are

$$u, v \geq 0, \quad u \leq \varphi^2 v, \quad v \leq \varphi^2 u. \quad (7)$$

Every three-vertex distance multiset is either $\{u, u, v\}$ or $\{u, v, v\}$. Hence all triangle inequalities are equivalent to

$$u, v \geq 0, \quad v \leq 2u, \quad u \leq 2v. \quad (8)$$

Because $2 < \varphi^2$, (8) implies (7). Conversely, every feasible semimetric must satisfy (8). Therefore (8) is the full invariant GL cone. Its only point on either coordinate axis is the origin, so a nonzero member has $u, v > 0$.

4 Optimization and equality cases

For (5), direct averaging over steps and ordered pairs gives

$$N_{a,b}(d) = \frac{au + bv}{S}, \quad D(d) = \frac{2(u + v)}{5},$$

so the objective is

$$R_{a,b}(u, v) = \frac{5(au + bv)}{2S(u + v)}. \quad (9)$$

Set $r = u/(u + v)$. Cone (8) is precisely $1/3 \leq r \leq 2/3$, and

$$\frac{au + bv}{u + v} = b + (a - b)r. \quad (10)$$

If $a > b$, the unique projective minimizer is $r = 1/3$, or $v = 2u$; if $b > a$, it is $r = 2/3$, or $u = 2v$; and if $a = b > 0$, (10) is constant throughout the nonzero cone. The respective values are exactly those in (1). The endpoint rays are attained by averaging the cut metrics of $\{0, 1\}$ and $\{0, 2\}$: their (u, v) pairs are respectively $(2/5, 4/5)$ and $(4/5, 2/5)$. Every cut metric is itself GL-feasible, via the embedding $x \mapsto \mathbf{1}_A(x) \in \mathbb{R}$. Hence the invariant lower bound equals the cut upper bound, completing the proof of Theorem 1, including both coordinate faces and the tie.

5 Exact finite verification and scope

The proof above is self-contained. A supplementary dependency-free Python script provides a separate exact check over the rationals: it enumerates all 30 nontrivial subsets, all 10 triples of vertices, and the two averaged endpoint cut metrics. It verifies the three cut signatures and multiplicities, the two triangle signatures, and the endpoint distances. This finite check is corroborative and is not substituted for any proof step. For an example of exact finite certificates and symmetry-reduced exhaustive enumeration in a different finite extremal problem, see Zijian Zeng et al. [2]; that work makes no claim about the Goemans–Linial relaxation or the theorem proved here.

The literature comparison in this note is deliberately bounded. The recent primary source identifies the one-weight cycle cases and a nearby character-based sufficient condition, but this note does not infer global novelty from the absence of the interior formula there.

Computational and AI assistance disclosure

An AI system assisted with proof organization, manuscript drafting, and consistency checks. The finite verification script uses only exact integer and rational arithmetic and exhausts the finite sets stated above. The mathematical proof is presented in full, independently of that computation; the author takes responsibility for the statements, calculations, citations, and final text.

References

- [1] Georgios Stamoulis. *Integrality Gap Bounds for the Goemans–Linial SDP on Finite Abelian Cayley Graphs*. arXiv:2609.05368v1. 2026. arXiv: 2609.05368 [math.CO].
- [2] Zijian Zeng et al. “Exact Finite Certificates for the Four-Dimensional Ternary Borsuk Problem and Five-Dimensional Ternary Kissing Codes”. In: *Preprints* (2026). DOI: 10.20944/preprints202608.1658.v1. URL: <https://doi.org/10.20944/preprints202608.1658.v1>.